Kubernetes Migration — Executive Summary

ShopFast E-Commerce Microservices Platform

Objective

Migrate ShopFast from Docker Compose to Kubernetes to gain elastic scaling, self-healing, standardized config/secrets, and automated GitOps delivery — with zero-downtime rollouts.

Current State

16 Spring Boot services, Eureka discovery, Keycloak identity, Kafka event backbone, PostgreSQL + Redis + Elasticsearch, ELK logging, Nginx edge, Compose orchestration.

Approach — 10 Phases

- 0. Cluster & tooling (minikube/kubectl/helm/skaffold)
- 1. Image prep (common-lib publish, SHA tags, JRE base)
- 2. Namespace + ConfigMaps/Secrets (Kustomize)
- 3. Stateful infra via Helm (Postgres, Redis, Kafka, ES, Keycloak)
- 4. Stateless services (Deployments, probes, HPA)
- 5. Ingress-Nginx + TLS + frontend routing
- 6. Observability (ELK + Prometheus/Grafana)
- 7. Resilience & security (Resilience4j, NetworkPolicies, SealedSecrets)
- 8. CI/CD + Argo CD GitOps
- 9. Local dev workflow (skaffold, port-forward)

Sequencing (low → high risk)

Scaffold config → deploy infra → Eureka + gateway → leaf services → order/cart/payment flow → frontend/ingress → observability → CI/CD/prod.

Expected Outcomes

- Automated rollouts/rollbacks and horizontal autoscaling
- Centralized, environment-specific config with sealed secrets
- Production-grade observability and resilience
- Reproducible, GitOps-driven deployments

Next step: implement Phases 2–4 (Kustomize base + reusable service template + infra Helm installs) as a working vertical slice.